

Green's Function for a Second-Order ODE with Heaviside Forcing

Source: Imperial College London, Carlo Contaldi, Differential Equations, Problem Sheet 9, Question 4.

Consider the following inhomogeneous ODE for $y(x)$:

$$\hat{L}[y] = y'' + 3y' + 2y = \Theta(x - x_0) \quad (0 \leq x \leq 1), \quad (1.1)$$

with homogeneous BCs $y(0) = y(1) = 0$, where $0 \leq x \leq 1$ and $\Theta(x)$ is the Heaviside (unit) step function.

1. Show that the Green's function for solving the above problem is

$$G(x, z) = \frac{e^{z-2x}}{e-1} \begin{cases} (e^x - e^0)(e^z - e^1) & x < z \\ (e^x - e^1)(e^z - e^0) & x > z \end{cases} \quad (1.2)$$

2. Show that this Green's function is *not* symmetrical.
3. Use the Green's function to find the solution to the above inhomogeneous problem (i.e. formed from the inhomogeneous ODE with the specified BCs), and show that it is

$$y(x) = \frac{e^{-2x}}{2(e-1)} [(e-1)(e^x - e^{x_0})^2 \Theta(x - x_0) - (e^x - 1)(e - e^{x_0})^2]. \quad (1.3)$$

Hint: Consider the cases $x < x_0$ and $x > x_0$ separately.

Solution. The homogeneous equation $y'' + 3y' + 2y = 0$ holds everywhere except $x = z$, with characteristic equation

$$\lambda^2 + 3\lambda + 2 = 0 \implies \lambda = -2, -1. \quad (1.4)$$

The Green's function therefore takes the piecewise form

$$G(x, z) = \begin{cases} A(z)e^{-2x} + B(z)e^{-x} & 0 \leq x < z \\ C(z)e^{-2x} + D(z)e^{-x} & z < x \leq 1. \end{cases} \quad (1.5)$$

Applying the homogeneous BCs $G(0, z) = G(1, z) = 0$:

$$A + B = 0, \quad Ce^{-2} + De^{-1} = 0 \implies B = -A, \quad C = -De. \quad (1.6)$$

Continuity of G at $x = z$ (substituting $B = -A$, $C = -De$):

$$Ae^{-2z} - Ae^{-z} = -Dee^{-2z} + De^{-z} \implies A = D \frac{e^{-z} - e^{-2z}}{e^{-2z} - e^{-z}}. \quad (1.7)$$

The jump condition $[\partial_x G]_{x=z^-}^{x=z^+} = 1$ (coefficient of y'' is unity) gives:

$$\underbrace{(2De^{-2z} - De^{-z})}_{G'(z^+)} - \underbrace{(-2Ae^{-2z} + Ae^{-z})}_{G'(z^-)} = 1. \quad (1.8)$$

Substituting (1.7) into (1.8) and expanding the denominator:

$$\begin{aligned} 2e e^{-4z} - e^{-3z} - 2e e^{-3z} + e^{-2z} + 2e^{-3z} - 2e e^{-4z} - e^{-2z} + e e^{-3z} \\ = e^{-3z} - e e^{-3z} = e^{-3z}(1 - e). \end{aligned} \quad (1.9)$$

Hence:

$$D = \frac{e^{-2z} - e^{-z}}{e^{-3z}(1 - e)} = e^z \frac{e^0 - e^z}{1 - e} = \frac{e^z(1 - e^z)}{1 - e}, \quad (1.10)$$

and from (1.7):

$$A = \frac{e^z}{1 - e}(e^z - e). \quad (1.11)$$

Assembling G for $x < z$ using $B = -A$:

$$\begin{aligned} G(x, z) &= A(e^{-2x} - e^{-x}) = \frac{e^z(e^z - e)}{1 - e}(e^{-2x} - e^{-x}) \\ &= \frac{e^{z-2x}}{1 - e}(e^z - e)(1 - e^x) = \frac{e^{z-2x}}{e - 1}(e^x - e^0)(e^z - e^1). \end{aligned} \quad (1.12)$$

Similarly, for $x > z$:

$$G(x, z) = \frac{e^{z-2x}}{e - 1}(e^x - e^1)(e^z - e^0). \quad (1.13)$$

Combining them, we find

$$\boxed{G(x, z) = \frac{e^{z-2x}}{e - 1} \begin{cases} (e^x - e^0)(e^z - e^1) & x < z \\ (e^x - e^1)(e^z - e^0) & x > z. \end{cases}} \quad (1.14)$$

It is rather clear that the Green's function we obtained does not have symmetry. This is expected, since the differential operator is not self-adjoint. To see this, consider the two equations:

$$\hat{L}_x G(x, z) = \delta(x - z), \quad \hat{L}_x G(x, z') = \delta(x - z'). \quad (1.15)$$

Multiplying the first by $G(x, z')$, the second by $G(x, z)$, subtracting and integrating:

$$\int_a^b \left[G(x, z') \hat{L}_x G(x, z) - G(x, z) \hat{L}_x G(x, z') \right] dx = G(z', z) - G(z, z'). \quad (1.16)$$

The left-hand side can be rewritten as:

$$\int_a^b G(x, z') [\hat{L}_x - \hat{L}_x^\dagger] G(x, z) dx + \text{boundary terms.} \quad (1.17)$$

If $\hat{L}$ is self-adjoint ($\hat{L} = \hat{L}^\dagger$) and the boundary terms vanish under homogeneous BCs, the left-hand side of (1.16) is zero, giving $G(z', z) = G(z, z')$.

For the operator $\hat{L} = \partial_x^2 + 3\partial_x + 2$, integrating by parts shows:

$$\hat{L}^\dagger = \partial_x^2 - 3\partial_x + 2 \neq \hat{L}, \quad (1.18)$$

since the odd-derivative term changes sign. Hence:

$$\hat{L} - \hat{L}^\dagger = 6\partial_x \neq 0, \quad (1.19)$$

and the left-hand side of (1.16) does not vanish.

Now, we are ready to obtain the particular solution to the differential equation:

$$y(x) = \int_0^1 dz G(x, z) \Theta(z - x_0). \quad (1.20)$$

For $x < x_0$, the discontinuity of the Green's function is to the left of x_0 , and thus the integral becomes

$$\begin{aligned} y(x < x_0) &= \int_{x_0}^1 dz G_{x < z}(x, z) = \frac{e^{-2x}(e^x - 1)}{e - 1} \left(-\frac{1}{2}e^2 - \frac{1}{2}e^{2x_0} + ee^{x_0} \right) \\ &= -\frac{e^{-2x}}{2(e - 1)}(e^x - 1)(e^{x_0} - e)^2. \end{aligned} \quad (1.21)$$

For $x > x_0$, the integral becomes

$$\begin{aligned} y(x > x_0) &= \int_{x_0}^x dz G_{x > z}(x, z) + \int_x^1 dz G_{x < z}(x, z) \\ &= y(x \rightarrow x_0^-) + \frac{e^{-2x}(e^x - e)}{e - 1} \left(\frac{1}{2}e^{2x_0} - e^{x_0} - \frac{1}{2}e^{2x} + e^x \right). \end{aligned} \quad (1.22)$$

Putting them together, we find

$$\boxed{y(x) = \frac{e^{-2x}}{2(e - 1)} [(e - 1)(e^x - e^{x_0})^2 \Theta(x - x_0) - (e^x - 1)(e - e^{x_0})^2]}. \quad (1.23)$$